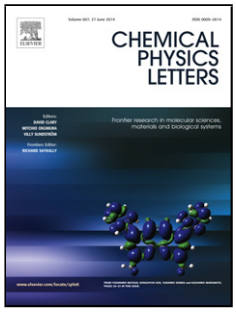
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Title: Connecting a new non-adiabatic vibrational mass to the bondingmechanism of LiH: a quantum superposition of ionic and covalent states

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PII: S0009-2614(15)00332-2
DOI: http://dx.doi.org/doi:10.1016/j.cplett.2015.04.062
Reference: CPLETT 32977

To appear in:

Received date: 30-1-2015
Revised date: 27-3-2015
Accepted date: 20-4-2015

Please cite this article as: Leonardo G. Diniz, Alexander Alijah, Ludwik Adamowicz, José R. Mohallem, Connecting a new non-adiabatic vibrational mass to the bondingmechanism of LiH: a quantum superposition of ionic and covalent states, *Chemical Physics Letters* (2015), http://dx.doi.org/10.1016/j.cplett.2015.04.062

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Highlights

1. Most accurate calculations of J=0 vibrational levels of LiH

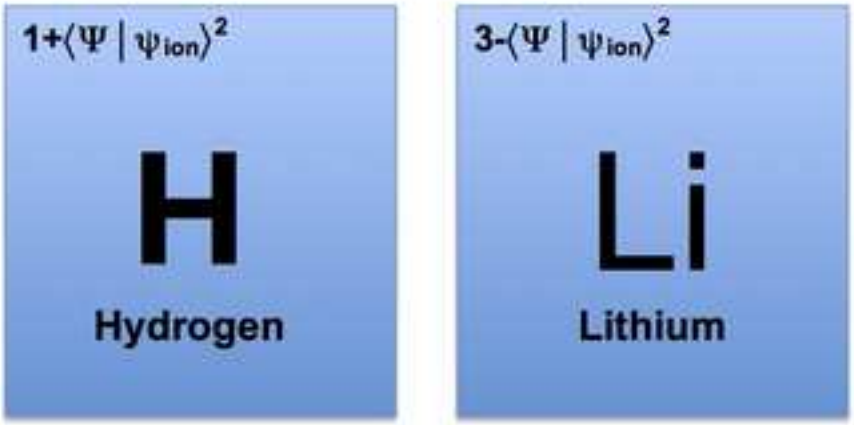
2. A new v=23 vibrational level

3. A valence-bond based effective nuclear reduced mass for non-adiabatic calculations

4. A new interpretation of the chemical bonding in LiH

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Connecting a new non-adiabatic vibrational mass to the bonding mechanism of LiH: a quantum superposition of ionic and covalent states

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Abstract

Non-adiabatic vibrational calculations performed with the accuracy of 0.2 cm^{-1} spanning the whole energy spectrum up to the dissociation limit for ^{7}LiH are reported. A so far unknown $v = 23$ energy level is predicted. The key feature of the approach used in the calculations is a valence-bond (VB) based procedure for determining the effective masses of the two vibrating atoms, which depend on the internuclear distance, R. It is found that all LiH electrons participate in the vibrational motion. The R-dependent masses are obtained from the analysis of the simple VB two-configuration ionic-covalent representation of the electronic wave function. These findings are consistent with an interpretation of the chemical bond in LiH as a quantum mechanical superposition of one-electron ionic and covalent states.

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I. INTRODUCTION

About 90 years ago, Heitler and London combined quantum mechanics and the variational method to explain, for the first time, how two H atoms form a chemical bond [1]. Since then the model of atomic electronic wave functions overlapping and leading to an enhancement of the electron density between them and formation of a bond became paradigm of the theory of chemical bonds. Adiabatic variational calculations have become the common tool to study the chemical bonding. However, if a crossing or pseudo-crossing of energy curves (surfaces for polyatomic systems) corresponding to different electronic states appears, the bonding electrons move between atoms in a way that a single adiabatic wave function is unable to describe near the crossing region (conical intersections for polyatomic molecules). An example of such a situation can be already found in the smallest polar molecule LiH. Qualitatively, the bond in LiH was explained by Mulliken [2] as a mixture of the ionic Li^+H^- and covalent $Li-H$ bonds. Following Mulliken the wave function of the ground electronic state of LiH is represented as a superposition two configurations, a ionic and a covalent one. The weights of these configurations are different at different bond lengths. Different analyses of the LiH electron density point either to a predominantly ionic [3] or predominantly covalent character of the bond [4, 5]. The truth seems to lie in the middle, i.e. near the equilibrium internuclear distance $Li-H$ is expected to have stronger ionic character and for the distances near dissociation the covalent character dominates, as the lowest dissociation channel of the molecule is $Li+H$.

In this work the bonding mechanism in LiH is used to determine an effective reduced mass of the system as a function of the internuclear distance. As the valence electrons shift along the bond when it shrinks and streches during the vibrational motion, the reduced mass changes slightly. This effect can be included in the calculations of the vibrational states. This is an effective way to account for a significant part of the so-called non-adiabatic effects. In the present calculations we implement this variable-mass approach in the calculations of the pure vibrational states (i.e. the $J = 0$ states) the LiH molecule.

LiH has been intensively investigated by both theorists and experimentalists (see Stwalley and Zemke [6]). Ten years ago, Coxon and Dickinson [7] analyzed the complete set of available experimental data consisting of 9084 spectral lines recorded for four LiH isotopologues, ^{7}LiH, ^{7}LiD, ^{6}LiH, and ^{6}LiD. Based on the analysis they generated effective potential energy

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curves (PECs) for the LiH $X^1\Sigma^+$ and $A^1\Sigma^+$ electronic states, which they used to compute vibrational energies. No more spectral data have been recorded since. A compilation of the pre-1986 experimental data by Chan et al. [8] is also used here for comparison.

For the theoretical characterization of the vibrational states two recent works are relevant to the results presented in this work. The first is that of Holka et al. [9] where a PEC calculated with very high accuracy and including the scalar relativistic, diagonal adiabatic, and non-adiabatic corrections was presented. The latter corrections were evaluated using an *ab initio* formalism by Bunker and Moss [10]. The second work is by Tung et al. [11] where all-electron explicitly correlated Gaussian functions with shifted centers were used to calculate the ground-state non-relativistic LiH PEC. The PEC was augmented by the diagonal adiabatic corrections calculated for ^{7}LiH, ^{7}LiD, ^{6}LiH, and ^{6}LiD and the four PEC are available from Ref. [11]. Tung et al. also calculated vibrational frequencies of all four isotopologues of LiH using the PECs augmented with diagonal adiabatic corrections. Though not explicitly mentioned in their paper, the atomic masses instead of nuclear masses were used in the calculations of the vibrational frequencies. In this way they effectively accounted for some part of the non-adiabatic effect. However, even with the high accuracy of the BO curves and with the diagonal adiabatic and relativistic corrections included in the PEC, the highest discrepancy of the vibrational term values, i.e. their energy with respect to the vibrational zero-point energy, was still of the order of 1 to 2 cm^{-1}, though the accuracy of predicted transition frequencies between neighbouring vibrational states is better than 0.3 cm^{-1} due to partial error compensation. The remaining error was attributed to non-adiabatic corrections not properly accounted for in the calculations.

Computation of the non-adiabatic corrections remains a true challenge to theoreticians. For the higher vibrational states, the avoided crossing of the PECs of the electronic ground state and the first excited state at around $R = 3\,\mathring{A}$ makes this challenge even harder to meet. This avoided crossing involves the diabatic curves corresponding to the ionic Li^+H^- and covalent $Li - H$ configurations [2]. The ionic character prevails near the equilibrium, but the covalent character becomes dominant when the bond length is stretched. For LiH, direct application of the non-adiabatic ab-initio formalism either through integration of the coupled differential equations or through alternative formulations, such as the ones suggested by Bunker and Moss [10] or by Herman and Asgharian [12], is still out of reach when spectroscopic accuracy is to be obtained. Note that the latter approaches incorporate the

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non-adiabatic effects in a single electronic state picture through vibrational and rotational reduced masses, which are different from each other and coordinate-dependent, and through a correction term added to the PEC. However, the authors in Ref. [9] concluded that the simple approach of using atomic masses gives smaller rms deviations from the experimental values than the not completely converged Bunker and Moss approach presented there.

An alternative approach to account for non-adiabatic effects is based on the realization that the source of these effects is due to a part of the electronic cloud moving with the nuclei when they vibrate or rotate [13–15]. This also leads to the concept of an effective coordinate-dependent reduced mass $\mu(R)$, which is connected to the perturbative second- and third-order nonadiabatic corrections [14]. Determining and using this mass not only makes the vibrational calculations more accurate, but helps unveiling the bonding mechanism in the molecule. In the next sections of the present work this approach is applied to LiH, for which, in view of the above cited change of character of the molecule with R, the search for an effective reduced mass $\mu(R)$ is particularly challenging.

In general, the present work could be related to the experimental techniques of femto-chemistry for observing some non-adiabatic effects on population transfer [16].

Finally, in order to better understand the connection of bonding mechanisms and the corrections introduced here, it is important to firmly establish the terminology used as well as the scope of the method: All concepts and terms refer to the Born-Huang theory of molecular structure (not to the Born-Oppenheimer-perturbation-theory). Truncating the Born-Huang series in just a single product function (one-state approximation) is referred to here as the adiabatic approximation. The corresponding PEC is obtained first at the fixed-nuclei Born-Oppenheimer approximation. Adiabatic corrections here mean the diagonal-Born-Oppenheimer-corrections (DBOC) to the PEC. Relativistic corrections, being mainly electronic, are empirically added to the PEC. Any further corrections are referred to as non-adiabatic (or non-diagonal) corrections, and are due to couplings with excited electronic states, either in the presence of avoided crossings or not. However, effective nuclear mass effects are expected to describe properly the couplings of the ground electronic state with other electronic states farther from avoided crossings. The actual suden change of electronic states, connected to avoided crossings, must still be treated in the context of diabatization. This point is further discussed in the last section. Atomic units are used throughout, except for non-adiabatic corrections, given in cm^{-1}.

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II. RESULTS

In the present work, a novel approach for determining the R-dependent effective reduced mass, $\mu(R)$, is introduced. The multi-configurational or multi-structure valence bond (MSVB) approach is used for this purpose.

A. The potential energy curve

In the vibrational calculations the BO PEC with adiabatic corrections from [11] augmented with relativistic corrections [9] is employed. Extending the PEC to very small distances is found necessary to obtain better numerical stability and well converged results particularly for highest vibrational states. The extension is done by fitting the ^{7}LiH PEC in the interval between 1.8 and 2.5 a_0 with function $f(R) = a + b\exp(cR)$, where $a = -8.0852$ E_h, $b = (7.8 \pm 0.4)$ E_h and $c = (-2.26 \pm 0.03)$ a_0^{-1}. Next the function is used to calculate some PEC points for $R < 1.8$ a_0.

B. The effective vibrational mass

Within the adiabatic approximation, the vibrational ($J = 0$) Hamiltonian of a diatomic molecule can be written as:

$$H = -\frac{\nabla_R^2}{2\mu} + W(R), \tag{1}$$

where μ is the nuclear reduced mass and $W(R)$ the PEC, which in the present case includes the BO energy, and the adiabatic and relativistic corrections. According to Bunker and Moss [10], if there is no avoided crossing, the non-adiabatic effects in calculating the vibrational energies can be effectively accounted for by using an R-dependent variable reduced mass, $\mu(R)$. The approach was applied by Holka et al. [9] but turned out to be difficult to use. Moreover, making the reduced mass independent on R, but replacing the nuclear masses by the atomic mass, as done in [11], was found to give better results.

Mohallem et al. [15] showed that the electron density in a diatomic molecule can be separated into three parts. The first is the density that moves along with the first nucleus when the molecule vibrates, the second part moves with the second nucleus, and the third part located in the bonding region does not move. This separation, which varies with the inter-

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nuclear distance was used by Mohallem et al. to determine an R-dependent reduced mass, $\mu(R)$. The moving parts were called *core* and the stationary part was called *valence*. A procedure based on the Mulliken population analysis was developed to determine the electronic masses of the core parts and of the valence part for simple homonuclear diatomics [15, 17] and for H_3^+ [18]. These applications produced R-dependent reduced masses, which, when used in the vibrational equation, resulted in a significant improvement of the results.

For LiH, the R-dependent reduced mass obtained using the approach based on the Mulliken population analysis does not lead to an improvement of the agreement between the experimental and calculated vibrational energies over the results obtained with the constant, atomic reduced mass. This may be related to the inability of this analysis to correctly describe the electronic charges which move with the nuclei and those which do not move when the molecule vibrates, in the situation where two configurations, the ionic (Li^+H^-) and the covalent ($Li - H$) one, mix in the wave function. Thus an alternative approach for determining $\mu(R)$ needs to be found and this is the goal of the present work. The proposed approach is based on the Valence-Bond (VB) method. For LiH the simplest VB wave function is a linear combination of the covalent and ionic configurations:

$$\Psi = C_{ion}\Phi_{ion} + C_{cov}\Phi_{cov}. \tag{2}$$

The statistical weights of these two configuration are given by:

$$W_{cov} = C_{cov}C_{ion}\langle\Phi_{cov}|\Phi_{ion}\rangle + C_{cov}^2\langle\Phi_{cov}|\Phi_{cov}\rangle \tag{3}$$

and

$$W_{ion} = C_{ion}C_{cov}\langle\Phi_{ion}|\Phi_{cov}\rangle + C_{ion}^2\langle\Phi_{ion}|\Phi_{ion}\rangle. \tag{4}$$

The VB2000 code [19] is used to perform the valence-bond-configuration-interaction (VBCI) calculations for LiH. In the VBCI method, the orbitals occupied by electrons in the VB configurations are atomic orbitals, i.e. each orbital is expanded in terms of the basis functions located at a single atomic center. The standard 6-31++G** basis set, the largest available in VB2000, is used in the VB calculations. As expected, the analysis of W_{ion} and W_{cov} shows domination of the ionic character of the VB wave function for shorter R's and the covalent character at longer R's. This behavior is illustrated in Figure 1 for non-orthogonal VB Φ_{ion} and Φ_{cov} configurations and for orthogonalized configurations obtained using the Löwdin method [20]. The two curves are negligibly different indicating that the usually

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non-orthogonal Φ_{ion} and Φ_{cov} have very small overlap, even at short bond lengths, which implies that the cross terms in Eqns. (3) and (4) are very small. The weights W_{ion} and W_{cov} are then the occupation probabilities of the covalent and ionic states, respectively. On this basis, the effective reduced mass, $\mu(R)$, can now be defined using the W_{ion} and W_{cov} probabilities as:

$$\mu_{eff}(R) = W_{ion}\mu_{ion} + W_{cov}\mu_{cov}, \tag{5}$$

where μ_{ion} and μ_{cov} are the R-independent covalent and ionic reduced masses. μ_{cov} is determined assuming that three electrons move with the lithium nucleus and one electron moves with the proton when the molecule vibrates. In calculating μ_{ion} the numbers of electrons are two and two, respectively. Note that in this model there are no immobilized electrons.

A R-dependent reduced mass can also be constructed using a core-valence approach as:

$$\mu'_{eff}(R) = \frac{(m_{\rm H} + \delta m_{\rm H})(m_{\rm Li} + \delta m_{\rm Li})}{m_{\rm H} + \delta m_{\rm H} + m_{\rm Li} + \delta m_{\rm Li}}, \tag{6}$$

where $m_{\rm H}$ is the proton mass, $m_{\rm Li}$ is the mass of the Li nucleus plus two electrons and $\delta m_{\rm H}$ and $\delta m_{\rm Li}$ are R-dependent mass increments defined as:

$$\delta m_{\rm H} = 2W_{ion} + W_{cov}, \quad \delta m_{\rm Li} = W_{cov}, \qquad (W_{ion} + W_{cov} = 1). \tag{7}$$

The mass increments, δm_H and δm_{Li}, are represented in the inset of Figure 1. They show the transition of the molecule from having ionic character at smaller R and having covalent character at larger R.

Interestingly, the two expressions for the reduced mass, Eqns. (5) and (6), differ only by a small term equal to $\mathcal{O}\left(\frac{1}{M}\right)$, where M is the total nuclear mass. Since, for more complex molecules, μ_{eff} is easier to calculate, it is used in the calculations performed in this work. Figure 2 shows a comparison of $\mu_{eff}(R)$ with μ_{ion} and μ_{cov}. The comparison shows that, as expected, $\mu_{eff}(R)$ converges to the covalent (or atomic) mass μ_{cov} at dissociation. In the inset, the VBCI ground and first excited states are shown, displaying the expected avoided crossing.

The vibrational states of ^{7}LiH are calculated with a renormalized Numerov [21] code updated to allow for μ to be R-dependent [22]. The numerical integration is performed within the range of $1.2\,a_0 \leq R \leq 30.0\,a_0$, and the parameters of the procedure are controlled to achieve numerical convergence at the level of $10^{-14}\,\mathrm{E_h}$.

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The differences between the vibrational frequencies calculated in this work and the experimental frequencies of Coxon and Dickinson [7] are presented in Figure (3). Up to about $v = 13$, the present results agree with both sets of experimental data [7, 8] to less than 0.2 cm^{-1}, an unprecedented accuracy among the theoretical calculations. It should be mentioned that the ab initio $\beta(R)$ correction from Eq. (3) in the work of Holka et al. was not fully evaluated [23]. The Sauer mass was simulated here from the vibrational g-factor data in Ref. [24]. Moving the g-factor in front of the kinetic energy operator to transform Eq. (4) of Ref. [25] into the form of Eq. (1) leads to a correction term to the potential curve whose inclusion alters the calculated vibrational energies by 10^{-2} cm^{-1} at most. In order to access the contribution from Gadea et al. [26], their non-adiabatic shifts were simulated by adding their vibronic shifts (E-E_{ac}) to the energies obtained here with nuclear masses and subtracting the results from the experimental energies. Since they resort to a diabatic (vibronic) approach, it is not easy to make comparison with effective mass based approaches, but it seems interesting to explore a connection if a future work. As the vibrational energies increase, the two sets of the experimental results start to disagree by about 0.5 cm^{-1}. In accordance, Chan et al showed that for $v > 13$ the average values of R lye already in the avoided crossing region or beyond [8], so that this feature can be the source of larger errors. On the other hand, recall that the measurements of higher states are usually less accurate and have, in consequence, smaller statistical weights in the fitting procedure like the one used in Ref. [7]. Also, note that the results obtained using the reduced mass determined from the atomic masses for higher v, where they should become more accurate, start to differ more from the experiment, but approach the present results. Thus, it is not clear how this increasing error is related to the accuracy of the present calculations.

Performing the calculations with a constant reduced mass averaged separately over each vibrational state (see [15, 17, 18]) yields practically the same results. Thus, the significant error reduction in the present calculations is solely due to the employment the VB-based procedure for calculating the effective reduced mass. Figure 3 also shows that among all the effective vibrational reduced masses considered so far for LiH, the constant reduced mass obtained from the atomic masses is the one which gives the best agreement with the experiments except for the R-dependent reduced mass of the present approach, which gives even better agreement.

Interestingly, the present calculations also reveal the existence of (unknown to both ex-

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periment and theory) the $v = 23$ bound vibrational state of ^{7}LiH, located only 8.03 cm^{-1} below the dissociation limit.

One may be curious if the present VB-based approach for determining $\mu_{eff}(R)$ can be extended to other diatomic molecules, particularly non-polar ones. To verify that, the procedure is applied to the H_2 molecule, for which essentially exact non-adiabatic calculations are available [27]. A VB calculation of H_2 involves three configurations, one covalent, $(H - H)$, and two ionic, (H^+H^- and H^-H^+). If all three configurations are used to calculate the effective reduced mass, one gets the same value of the mass as obtained using the atomic masses regardless of the length of the bond. If, however only the covalent structure is used to determine the core masses, the agreement with Wolniewicz's calculations [27] becomes remarkably good, within 0.1 cm^{-1} for v up to 6, as shown in Figure 4. Note that this level of agreement is difficult to reach without resorting to a scaling procedure like the one used before [17]. Thus again, the reduced mass determined using the VB-based procedure allows for a better representation of the non-adiabatic effects in H_2 than other effective-nuclear-mass procedures. More rigorous derivations of effective reduced masses have been presented [13, 14, 25], but these formulations are difficult to transfer to other molecules. Disagreement of the present results from the exact ones is observed for $v > 5$, which reaches its maximum around $v = 10$. It could be related to a possible inability of the simple VB wavefunction used to deal with the immobilized electrons for these states, a problem that does not exist for LiH, since the two contributing configurations have definite and simple distribution of the electron charge. For H_2 may be other covalent states which cross the ionic one should be included, but this procedure is not straightforward in the present contest.

C. Discussion: the bonding as a superposition of one-electron states

Kutzelnigg used simple LCAO calculations and determined that for H_2^+ and H_2 about 60% of the electrons participate in the ground-state vibrational motion along with the nuclei [13]. Thus, according to the present model, the remaining 40% accounts for the valence electrons, which are located on the bond and does not move when the molecule vibrates. Similar percentages were obtained by considering as core electrons the diagonal atomic contributions from Mulliken's population analysis [15, 17]. However, as shown in [15], to match the exact results, the percentage of the core electrons for the low-lying vibrational states should be

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about 80% and increases up to 100% for higher states. The present VB-based procedure gives this 80% without resorting to any empirical scaling.

LiH is a typical ionic diatomic molecule. At and around the equilibrium bond length the molecule is ionic and it is primarily bound by an electrostatic attraction of two atomic ions. Thus, there is no electron density built up in the middle of the molecule. At the point where the molecule becomes covalent, the atoms are already separated far enough so again there is little electron density in the middle of the bond. This explains why all electrons in LiH move when the molecule undergoes a vibrational motion. As the present calculations show, knowing the weights of the ionic and covalent configurations which are used to determine the R-dependent reduced mass, is key to obtain good agreement with the experiment concerning the vibrational energies. As the VB ionic and covalent configurations practically do not overlap (see Figure 1), the VB atomic orbitals do not overlap either. This indicates that all electrons only occupy quasi-orthogonal atomic orbitals (not shared two-center orbitals). Thus all electrons are atomic core electrons and all are involved in the vibrational motion in all vibrational states. The quasi-orthogonality, even at small bond lengths, of the two configurations and hence of the atomic orbitals of Li and H is consistent with a probabilistic interpretation of the chemical bond in LiH where the transition from ionic to covalent behavior is determined by R-dependent probabilities of one electron occupying either Φ_{ion} or Φ_{cov}. Thus, an analysis of the vibrational spectrum of LiH unveils the nature of bonding in this molecule.

Lack of orbital overlap does not mean, however, that $\mu(R)$ of Eq. (5) is R-independent. Rather, the bonding electron is "divided" between the two nuclei, incrementing their masses by δm_H ($\geq$ 1) and δm_{Li} ($\leq$ 1) according to the probabilities given by W_{ion} and W_{cov}, respectively. This is shown in Figure 1. Hence, the picture which emerges from the above analysis is that of a loose electron occupying the orbitals of the two atoms with measurable probabilities, without passing through a valence state. The transition from ionic to covalent behavior is determined by R-dependent probabilities of one electron occupying either Φ_{ion} or Φ_{cov}.

It might surprise that a simple two-configuration VBCI approach considered here is able to produce a good reduced mass for the vibrational calculations of LiH. It is certainly not suitable to compute the PEC with high-accuracy, but seems to identify correctly the fraction of the electron density which moves with each nucleus. It must be made clear at

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this point that the electron jumping model used here applies just to the building of the effective mass, that is, despite its appealing physical interpretation it does not apply fully to pseudo-crossing effects where the actual electron jumping takes place. It just handles the the further couplings with electronic states, which can be seen as the electron dragging by nuclei. However, in view of the weak nature of the avoided crossing involving the ground state of LiH, it seems also reasonable to consider that the approach diminishes the influence of the avoided crossing (where the energies become vibronic) on the calculated vibrational energies.

Finally, it is interesting to compare the present model of the LiH bonding with the model of Fantuzzi and Nascimento according to which the bond in LiH is predominantly covalent [5]. They suggested that the binding is explained by a semi-classical ionic effect that moves the electronic density from Li to H and an opposing quantum interference effect (the superposition of atomic orbitals) which moves the electronic density towards the bonding region. According to Fantuzzi and Nascimento, the quantum effect is more important near the minimum of the adiabatic PEC so the bond has mainly a covalent character. The conclusion of Fantuzzi and Nascimento that the ionic contribution moves the density from one atom to the other atom is consistent with our results. What is found different here is that the same occurs for the quantum contribution— there is no electron fraction immobilized between the atoms and a single electron is responsible for the bond. This single electron is in an approximate superposition of ionic and covalent states. That is, the present model disagrees with that of Fantuzzi and Nascimento in their interpretation of the quantum effects as being representative of electron sharing. In this sense the two versions are not necessarily contradictory, but the present model establishes the quantum component of the bonding mechanism on different physical grounds.

III. CONCLUSIONS

The model presented in this work describes the mechanism by which fractional amounts of the electrons in the LiH molecule change their distribution among the atomic centers. This involves one electron transition between the ionic and covalent states. The process can be described by a simple two-configuration VB calculation where the wave function is a superposition of ionic and covalent configurations. It is found that all electrons in

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LiH participate in the vibrational motion and there are virtually no electrons which are positioned in the middle of the bond and are immobile. The vibrational frequencies obtained in the present calculations with the R-dependent effective reduced mass determined with a procedure based on the VB calculations agree very well with the experimental values.

The idea of a panel for re-discussion of the concept of valence was advocated recently [28]. The present results provide yet another example why such a panel would be highly desirable and timely.

Acknowledgements

We acknowledge CNPq and Capes for a financial support (PEV grant 401846/2013-0) and Profs. R. Vianna, A. Cesar and V. Tyuterev for useful discussions.

IV. FIGURE CAPTIONS

Figure 1: Statistical weights of the valence bond structures of LiH. Mull: Mulliken weights, Löw: Löwdin weights. The insert figure shows the mass increments δm_H and δm_{Li} in a.u.

Figure 2: The VB reduced mass $\mu_{eff}(R)$ compared to μ_{ion} and μ_{cov} (see definitions in text). The insert figure shows the two low-lying VBCI adiabatic curves and the avoided crossing.

Figure 3: Difference between calculated and experimentally derived (Coxon and Dickinson [7]) term values ($v \to 0$) for ^{7}LiH. NU: nuclear mass, Holka: from Holka et al. [9], Sauer: from Sauer et al. [24], Gadea: from Gadea et al. [26], AT: atomic mass, MSVB: valence bond, this work.

Figure 4: Absolute non-adiabatic correction for H_2. Exact: from [27], AT: (atomic mass) and Cov: this work.

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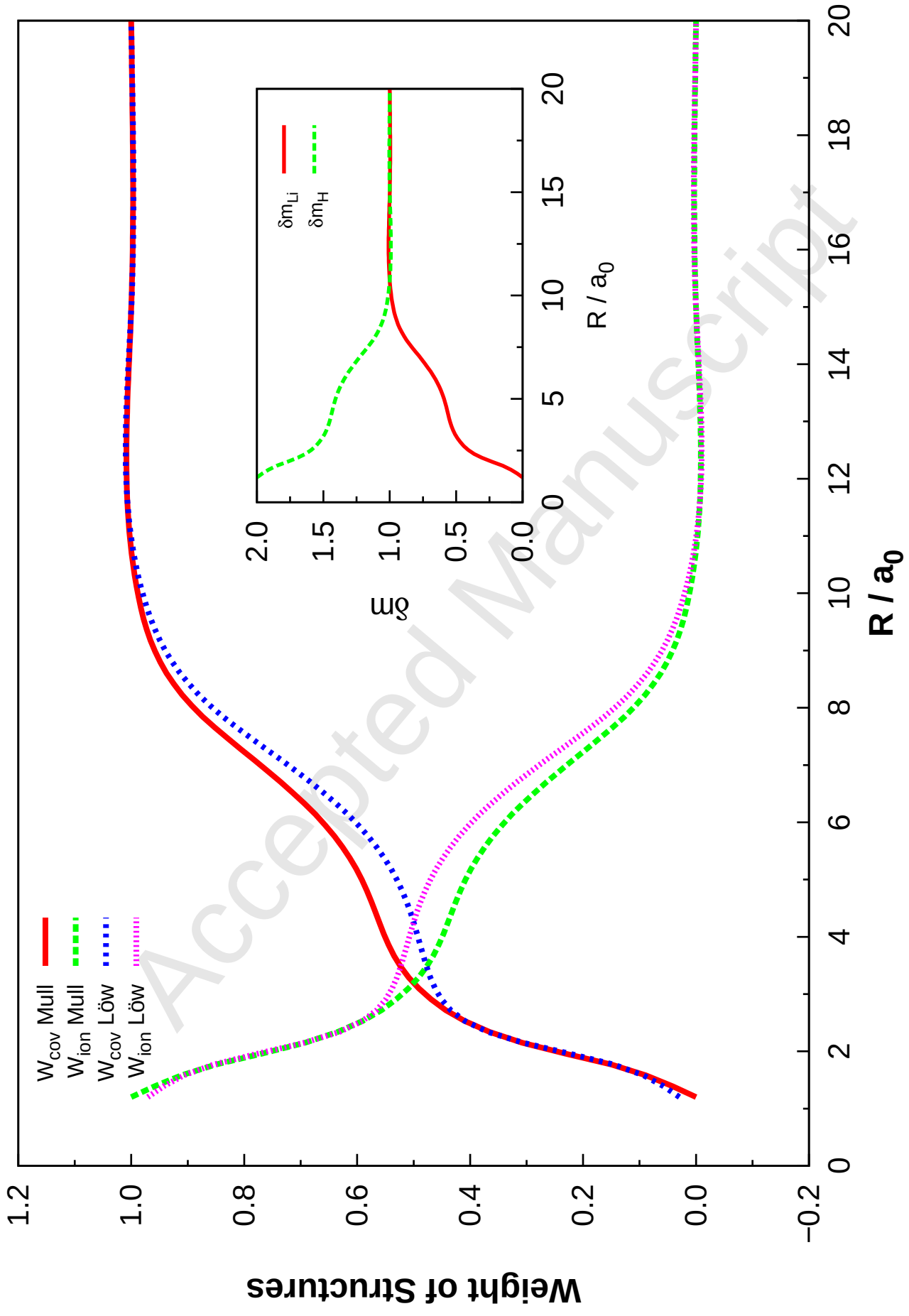
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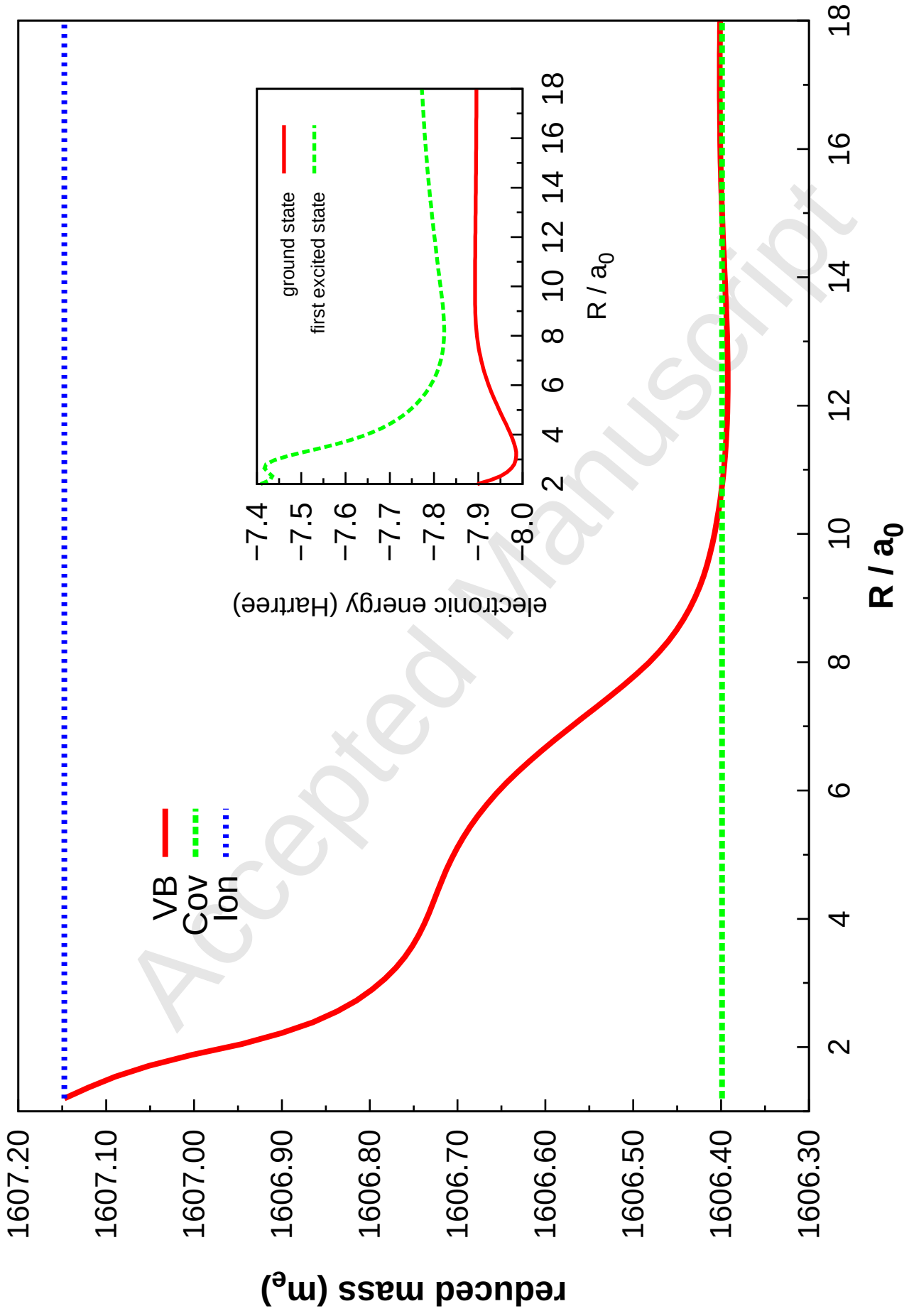
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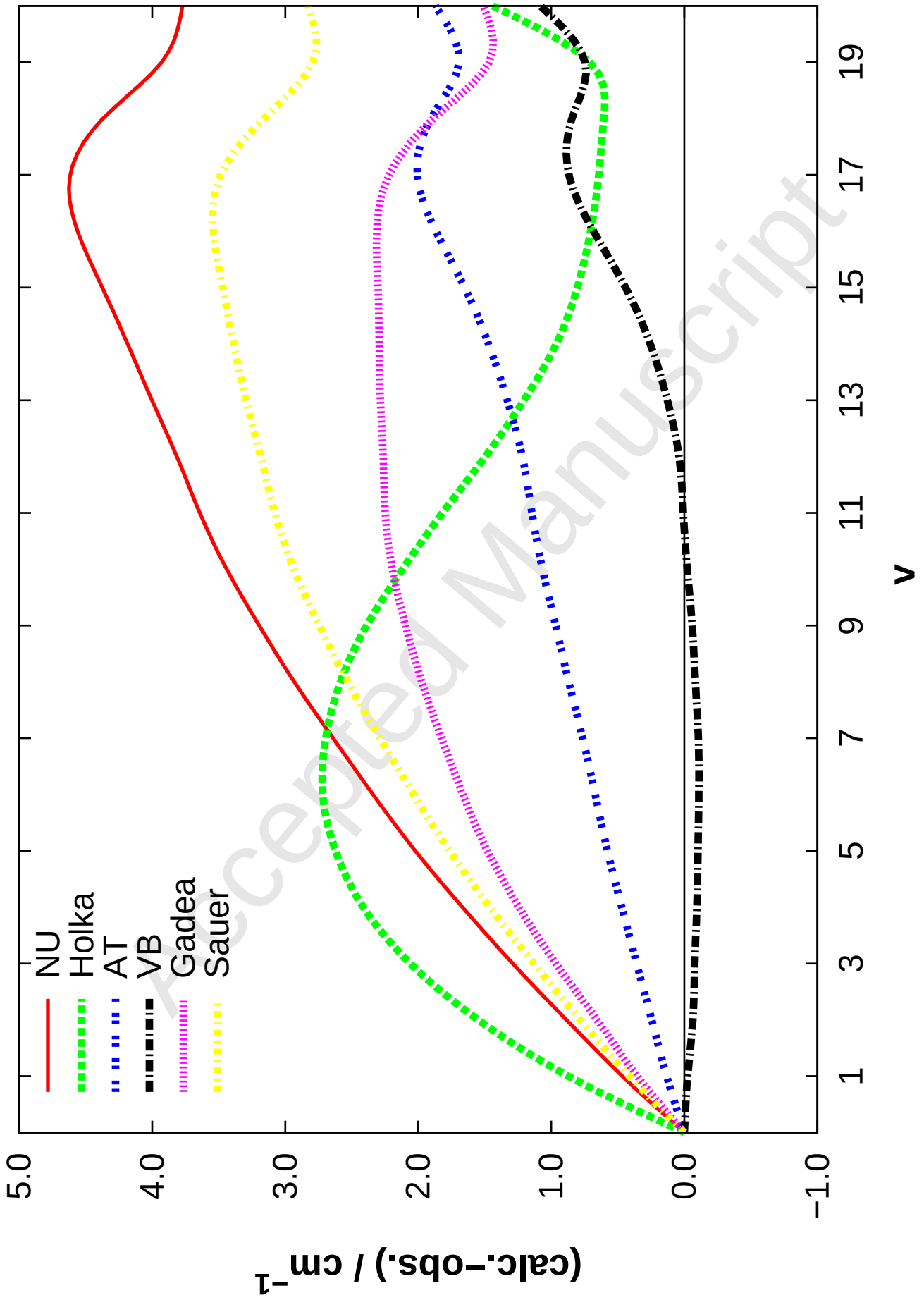
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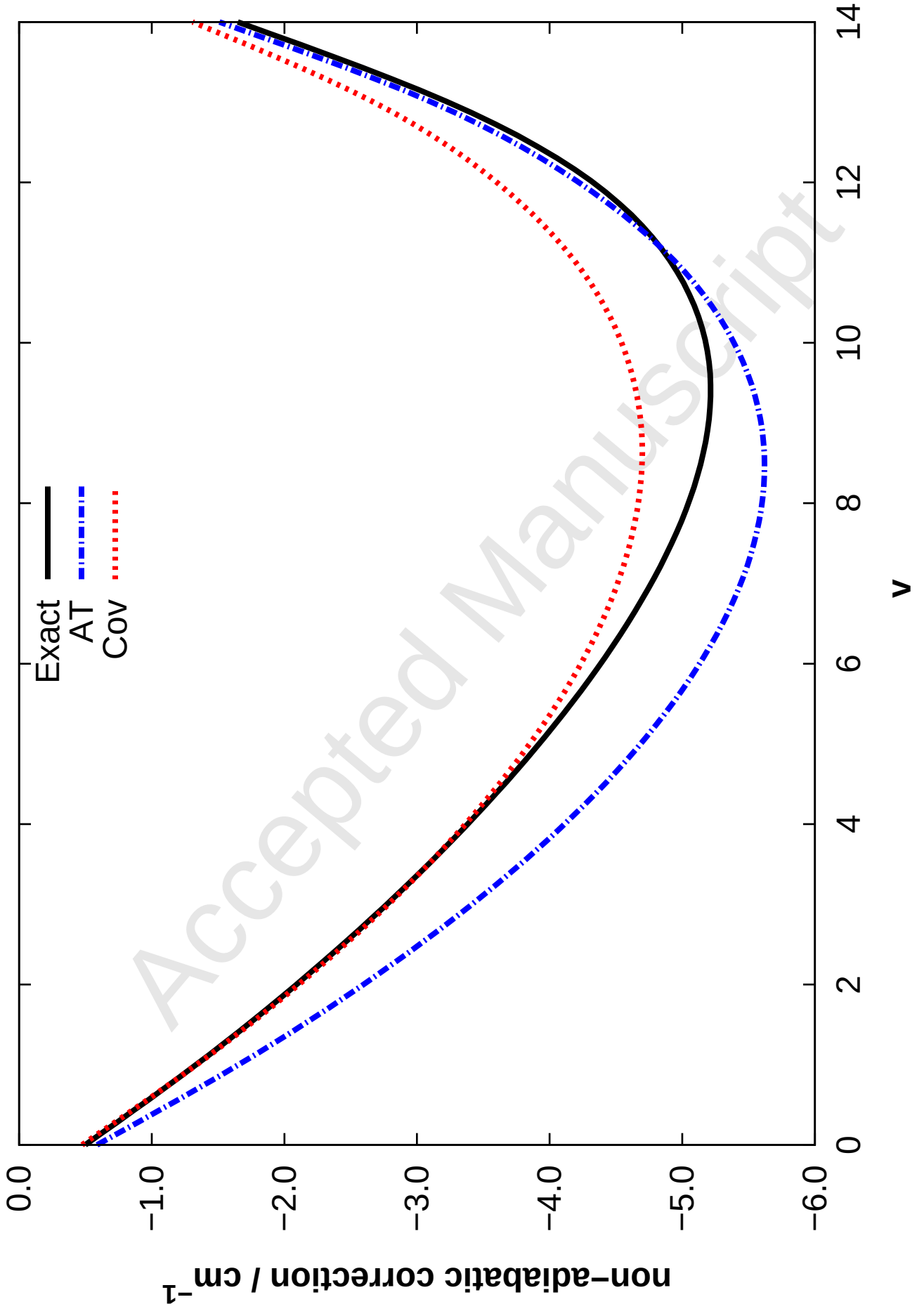
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